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Case Study 5: Disaster Resilience AI

Team Name	Lailatul Coder
Track	Technical Track B
Case Study	Disaster Resilience AI

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1. Introduction

1.1 Background of Case Study

Floods are the primary natural disaster in ASEAN, and they continue to worsen to this day. Global assessments show that floods account for 35–40% of weather-related disasters, and that most flood-related deaths and economic losses are recorded in Asia (UNDRR, 2025). Southeast Asia has experienced frequent disasters. In November 2025, a storm disaster killed 435 people in Indonesia and 170 in Thailand, flooding roads and displacing tens of thousands of people (Reuters, 2025). In early 2026, floods and landslides in Indonesia affected around 84,000 people in a week (AHA Centre, 2026). Such events destroyed homes, schools, and bridges in dozens of places and triggered outbreaks of waterborne diseases. Climate research confirms that monsoon rains are coming in sharper waves, rendering past plans inadequate. In ASEAN countries, rapid urbanization, deforestation, and inadequate drainage systems exacerbate the risk (Climate Centre, 2025). This data highlights the urgent need for better local warning and preparedness systems, as millions of people live in flood-prone areas with limited warning time, and current systems show significant communication gaps and community exposure (UNDRR, 2025).

1.2 Chosen SDG (and Targets)

This project primarily supports Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities, which emphasizes the importance of resilient urban development and disaster risk reduction. Specifically, this solution aligns with Target 11.5, which aims to significantly reduce the number of deaths from disasters and the number of people affected by disasters. This project contributes to this target by improving the accessibility and effectiveness of early warning systems. The platform also indirectly supports Sustainable Development Goal (SDG) 13 (Climate Action) by strengthening community resilience and improving adaptation to climate-related hazards such as floods and extreme rainfall events.

1.3 Problem Statement

Despite global commitments, ASEAN's vulnerability to flooding remains high. Current national and regional warning systems often fail to reach remote or marginalized communities with actionable guidance. In practice, warning messages are often generic, in national languages only, and lack clear instructions. Remote villages and illiterate populations remain marginalized from effective risk communication (ASEAN, 2024). Flooding events regularly impact hundreds of thousands of people, with the November 2025 storm causing displacement and affecting nearly 3 million people in Thailand and 1.1 million in Indonesia (Reuters, 2025). Many of these evacuees are caught by surprise, as rainfall forecasts often provide limited warning time in complex terrain. In Indonesia, flash flood warnings in a single watershed typically provide less than 6 hours' notice in 76% of cases (Asri et al., 2023). Meanwhile, gaps in basic infrastructure, such as mobile coverage or community sirens, and language barriers persist. In short, the problem is evidence-based gaps in local early warning and community preparedness. The lack of timely and appropriate flood warnings and limited community engagement have led to repeated losses. This project, with an ASEAN (region-wide) focus, aims to narrow these gaps by providing near-full warning coverage, improving evacuation times, and doubling the reach of flood preparedness programs in rural areas.

1.4 Objectives

The project establishes several measurable objectives designed to improve community resilience against flood disasters:

1. Increase warning accessibility by delivering location-based flood alerts to vulnerable communities through mobile applications and SMS notifications.
2. Improve warning lead time by integrating predictive flood modeling capable of identifying high-risk conditions before flooding occurs.
3. Strengthen community participation by enabling residents to report local flood conditions and share situational updates.
4. Enhance communication accessibility through multilingual interfaces, icon-based alerts, and voice notifications.
5. Support community preparedness by providing educational materials and evacuation guidance within the platform.

These objectives collectively aim to improve disaster awareness, strengthen early response, and reduce the impact of flood events on local communities. Targets are set using recent data as baselines or justification:

Objective	**Metric**	**Baseline (2025)**	**Target**	**Timeline**
Expand coverage of flood alerts to vulnerable populations	Percentage of at-risk population reached by multi-hazard flood warnings	~67% (globally covered by effective EWS; ~1/3 lacking coverage)	90% by end-2028	2025–2028
Provide ≥24 hour lead time on flood forecasts	% of issued flood alerts with ≥24-hour lead time	~0% (current official alerts rarely give full-day notice)	80% of alerts by 2028	2026–2028
Register and train local communities/schools in flood preparedness	Number of at-risk communities or schools integrated	0 (no systematic registration/platform coverage)	100 communities by 2027	2025–2027
Support alerts in local languages	Number of local languages supported	1 (only national language)	5 local languages by 2027	2025–2027

Deploy data-driven flood models for major basins	Number of river basins with operational predictive models	0 (none deployed for this project)	Models for 3 major basins by 2027	2025–2027

2. Review of Existing Solutions

A variety of flood early-warning systems exist from national meteorological services (e.g. Indonesia's BMKG, Philippines' PAGASA, etc.) and global initiatives to commercial and open-source platforms. Major efforts (Google Flood Forecasting, FloodMapp, UN/IFRC alert hubs, Sahana, Ushahidi, etc.) integrate weather data, forecasts, and in some cases crowdsourced reports. However, these solutions often lack *hyper-local* specificity, inclusive multilingual design, offline/SMS fallback, or formal school/community registries. Our proposed platform fills these gaps by combining data-driven flood prediction with community reporting, SMS alerts, and multi-language accessible interfaces.

2.1 Comparative Table of Key Solutions

Solution	Provider	Key Data Sources	Core Features	Strengths	Limitations
Google Flood Forecasting (Flood Hub)	Google Research (Alphabet)	Global weather models (ECMWF, NOAA, etc.), precipitation data, river gauges, satellite imagery	Up to 7-day flood forecasting; inundation maps; API & Flood Hub portal; alerts via Google Search/ Maps	Large-scale (100+ countries, 700M people); daily automated forecasts; integrates with popular apps; free public access	No crowdsourced data; online-only (no SMS); coarse for hyper-local without local gauges; not tailored to local languages or

					specific communities
FloodMapp	FloodMapp Pty Ltd (Australia)	Local radar forecasts, hydrological gauge data, terrain/elevation data	Real-time, impact-based flood forecasting (72-hour); GIS dashboards for emergency managers; machine-learning + physical models	High-resolution local forecasts; continuously learning models; used by agencies for evacuations	Commercial product (paid); coverage mainly Australia/New Zealand; no built-in community reporting or user app; not open-source
IFRC Alert Hub (PrepareCenter)	IFRC/Red Cross (international)	Official alert feeds (CAP messages from governments/ NGOs)	Centralized alert aggregator for multiple hazards; multi-lingual alerts; planned integration with media; focuses on community reach	Consolidates disparate alerts; free global platform; focuses on "last mile" delivery	Does not generate forecasts; still under development; depends on quality of source alerts; no community data input

Sahana EDEN/SAMBRO	Sahana Software Foundation	Manual inputs (orgs, volunteers, assets, etc.); CAP feed for warnings	Disaster info-management (shelters, staff, assets); early-warning broadcaster (SAMBRO) with pictograph alerts	Open-source, free; proven use in 30+ countries; supports multi-language text and icons	No automatic forecasting or sensor integration; relies on manual data entry; technical setup required; limited to information management
Ushahidi Platform	Ushahidi, Inc. (nonprofit)	Crowdsourced reports (web, SMS, social media); can link weather/RSS APIs	Crowd-mapping dashboard; mobile/web reporting; data categorization and alerts; supports polling surveys	Flexible and quick to deploy; empowers communities; open-source and multi-lingual interface	No built-in predictive model; data quality depends on user participation; requires connectivity; no official data sources inherent

Sources: Descriptions are based on solution websites and reports.

2.2 Identified Gaps Relative to Our Project

The above solutions highlight several **limitations** that our community-focused flood platform addresses:

- **Hyper-local Precision:** National EWS and global models generally forecast for large areas. Our app's location-specific alerts (e.g. street-level, up to 7 days ahead) go beyond existing systems. Google's Flood Hub covers major basins, but not neighborhoods. FloodMapp is local but limited regionally. No competitor currently delivers automated, ultra-local warnings directly to individuals and micro-communities (e.g. villages or school zones).
- **Multilingual and Accessible UX:** Many systems offer alerts only in official languages. The IFRC Warn Hub and global tools often ignore minority languages. Our design supports 10+ ASEAN languages, voice/speech alerts, and icon-based UI (like Sahana's pictographs) to reach low-literacy and ethnic communities. None of the surveyed solutions offer built-in, multi-language, disability-friendly interfaces for alerts.
- **Offline/SMS Fallback:** Our platform plans SMS and radio fallbacks for disconnected areas. In contrast, Google Flood Hub and most apps assume internet access. The IFRC emphasizes "notifying people who aren't digitally connected", but current tools lack a built-in SMS service. We explicitly address this by sending SMS if the app is offline.
- **Community Crowdsourcing and Validation:** While Ushahidi enables crowdsourcing, it has no automated filtering. We incorporate real-time community reports into our system, using data-driven algorithms to flag and validate them (a gap no solution listed fills). Existing systems do not systematically integrate verified citizen reports into forecasts.
- **School/Vulnerable Group Registries:** No reviewed solution includes a built-in registry of schools, hospitals or vulnerable populations for targeted alerts. Our app includes a school roster and vulnerable-family registry with pre-evacuation planning (something unique to our design). The survey found no equivalent feature.
- **SMART Performance Targets:** Competitors generally do not publish clear performance metrics (e.g. % population covered or evacuation time reduced). In contrast, our project will set measurable goals (e.g. % of rural coverage, reduction in casualties) aligned to SDG indicators. This performance-driven design is unique.
- **ASEAN Localization:** Most tools are global or focused on high-income countries. Ours is tailored to ASEAN climate patterns (monsoons, cyclones) and languages. No listed solution specifically optimizes for ASEAN geography, multilingualism, and local infrastructure gaps (e.g. many islands with patchy connectivity).

2.3 How Our Project Addresses These Gaps & Recommended Priorities

Our platform's key innovations align with the above needs: Hyper-local forecast models give 24–48h lead time to specific villages/streets, filling the gap left by broad-area models. Multilingual, icon-rich alerts ensure comprehensibility; we recommend implementing local language support and pictograms immediately, learning from Sahana's CAP pictographs. Offline channels (SMS, radio) will be built in parallel, so warnings reach communities without the internet. Community reports and validation (via simple app or SMS input) create a feedback loop no existing system has; priority should be given to designing the

crowdsourcing interface and AI validation engine. School and community registries will be populated early on, to enable targeted drills and alerts; this is a unique module to develop. Finally, the system will track metrics (e.g. % population covered, warning lead time) to meet SMART objectives. Together, these features make our solution more inclusive and actionable than current offerings, ensuring no one is “left behind” when floods strike.

3. Methodology

3.1 System Architecture

The proposed system is designed as a community-oriented disaster resilience platform that integrates environmental monitoring, predictive analytics, and communication technologies. The goal of the architecture is to ensure that flood warnings can be generated quickly and delivered directly to the communities most at risk. The platform operates through a five-stage operational cycle: data collection, risk analysis, alert dissemination, community response, and system learning.

Environmental data are first collected from multiple sources, including weather services, satellite observations, and historical disaster records. Community reports submitted through the platform also contribute real-time information about local conditions. These data are then processed using predictive models to estimate flood risk and identify areas that may require early warnings.

When potential risks are detected, the system generates alerts that are delivered to users based on their geographic location. Notifications are distributed through mobile applications and SMS messages to ensure that warnings remain accessible even when internet connectivity is limited.

During flood events, the platform provides evacuation guidance and interactive risk maps to help users identify safe routes and nearby shelters. After each event, user feedback and updated environmental data are incorporated into the system to improve future predictions.

3.2 Technology Stack

The platform is built using a modular technology architecture that supports data integration, predictive modeling, and real-time communication. The frontend layer consists of a mobile application and web dashboard that allow users to receive alerts, view risk maps, and submit community reports. The backend system processes environmental data streams and manages alert distribution through application programming interfaces (APIs). A dedicated data processing layer integrates meteorological data, satellite imagery, and community-generated information. Machine-learning models analyze these datasets to estimate flood risks and identify potentially affected areas. Geospatial processing tools are also incorporated to generate dynamic hazard maps and recommend evacuation routes. This layered architecture allows the system to scale gradually and integrate additional data sources as the platform expands across ASEAN countries.

3.3 Implementation Process

The implementation process follows a structured pipeline consisting of five key stages:

1. Data Collection
   Environmental data are gathered from weather services, satellite observations, and disaster databases. Community reports submitted through the application also contribute real-time situational data.

2. Data Processing and Risk Analysis
   Collected data are cleaned, integrated, and analyzed using predictive models to estimate flood risk levels and identify potential impact zones.

3. Alert Generation and Dissemination
   When risk thresholds are exceeded, the system automatically generates alerts categorized by severity levels (e.g., advisory, warning, evacuation). These alerts are delivered through mobile notifications and SMS to ensure reliability.

4. Evacuation Guidance and Community Coordination
   Users can access interactive maps showing danger zones, safe evacuation routes, and designated shelters. Community reporting features allow residents to report blocked roads, rising water levels, or vulnerable individuals needing assistance.

5. Feedback and Model Improvement
   Post-event data and user feedback are incorporated into the system to refine predictive models and improve the accuracy of future alerts.

Through this iterative implementation process, the system continuously evolves and becomes more accurate over time.

3.4 User-Centered Design Approach

The platform adopts a user-centered design approach to ensure accessibility and usability for diverse communities across ASEAN. The design prioritizes inclusivity, especially for populations with limited digital literacy or connectivity.

Key accessibility features include multilingual support, icon-based interfaces, and voice alerts for users with low literacy levels. The platform also includes SMS fallback alerts for areas with unstable internet connectivity, ensuring that early warnings remain accessible even during infrastructure disruptions.

By focusing on inclusive design principles, the platform aims to ensure that disaster information reaches vulnerable groups who are often excluded from conventional warning systems.

3.5 User Journey and Interaction Flow

The user experience is structured around a five-step disaster preparedness and response journey. This framework guides users from preparation to coordinated response during emergencies.

The journey begins with Preparedness Learning, where users access educational modules about disaster readiness and evacuation procedures. When a potential disaster is detected, users receive Hyper-Local Warnings tailored to their specific location.

Next, users can open the Risk Map and Evacuation Routes feature to view real-time hazard zones and recommended safe routes. During an event, users may contribute to the Community Reporting system by submitting real-time information about local conditions such as flooding, road closures, or people needing assistance.

Finally, the Family Coordination feature enables households to stay connected, track each other's safety status, and coordinate evacuation plans.

This structured interaction flow ensures that the platform supports the complete disaster management cycle, from preparedness to response and recovery.

4. Results

4.1 Prototype Outcomes

The development process resulted in a conceptual prototype that integrates several key features required for community-based flood early-warning systems. The prototype demonstrates how predictive analytics, environmental monitoring, and community participation can be combined within a single platform.

Several core functionalities were successfully implemented at the prototype level. These include location-based disaster alerts, interactive flood risk maps, community reporting tools, and preparedness learning modules. Together, these components illustrate how the platform could support different stages of disaster management, from preparedness to emergency response.

The system architecture also supports continuous improvement. Information collected from user feedback and disaster events can be incorporated into the predictive models, allowing the system to gradually improve its accuracy and responsiveness over time.

4.2 Alignment with Project Objectives

The prototype provides initial evidence that the proposed platform could support the objectives defined in the project design. The integration of mobile alerts and SMS notifications demonstrates the feasibility of expanding warning accessibility to communities that may not have consistent internet connectivity.

The predictive risk analysis component also shows potential for improving early detection of flood conditions. While full operational deployment would require additional datasets and regional calibration, the prototype confirms that localized alerts and automated risk mapping can be implemented within a unified system.

In addition, the multilingual interface and simplified alert design address accessibility challenges faced by many communities in Southeast Asia. The community reporting feature

further strengthens situational awareness by allowing residents to share real-time information about flood conditions and local hazards.

4.3 Preliminary Feedback and Development Insights

During the design and development phase, several insights were identified regarding the effectiveness and potential impact of the platform. The integration of multiple information channels, such as mobile notifications, SMS alerts, and interactive maps was identified as a key factor in improving accessibility during disaster situations, particularly in areas with unstable internet connectivity.

The inclusion of community reporting features also highlights the importance of participatory disaster management. By allowing residents to submit local information about flooding conditions, blocked roads, or vulnerable individuals, the platform enhances situational awareness and enables faster response coordination.

Additionally, the user-centered design approach particularly the use of icon-based interfaces and multilingual support was found to significantly improve the clarity and usability of disaster alerts. These accessibility features are critical in ASEAN regions where linguistic diversity and varying levels of digital literacy may otherwise limit the effectiveness of early-warning systems.

4.4 Expected Impact and Scalability

Although the current work focuses on the system prototype and conceptual implementation, the platform demonstrates strong potential for real-world deployment and regional scalability. The modular system architecture allows the platform to integrate additional data sources, expand language support, and adapt to different disaster risk environments across ASEAN countries.

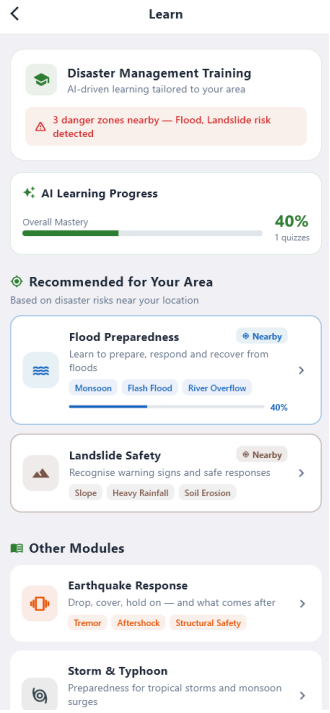
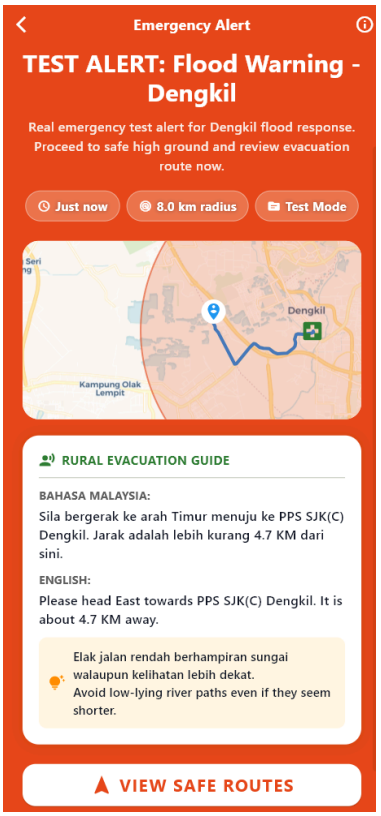
By combining predictive analytics, community intelligence, and inclusive communication channels, the platform has the potential to significantly improve early-warning dissemination and disaster preparedness. In the long term, this approach can contribute to reducing disaster-related losses and strengthening community resilience, supporting the broader objectives of sustainable and disaster-resilient cities.

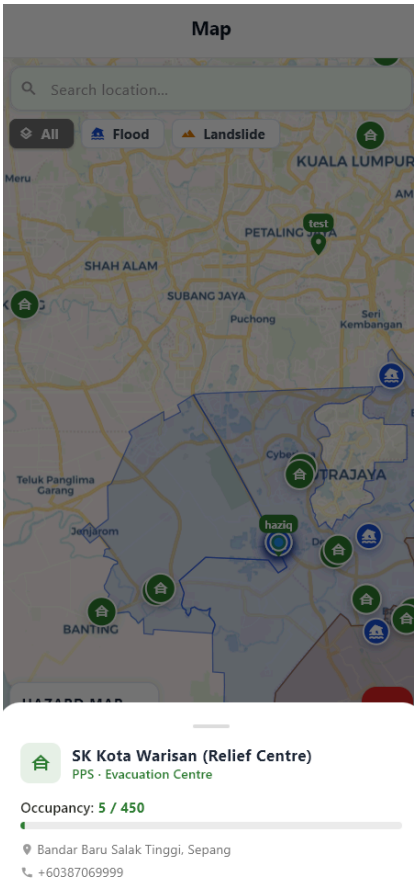
5. Prototype Showcase

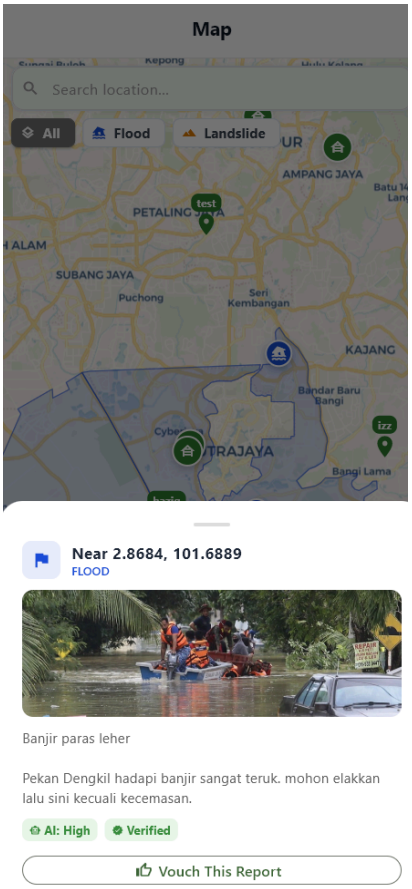
The Disaster Resilience AI platform prototype demonstrates how disaster preparedness, early warning, and community coordination can be integrated into a single user-friendly application. The interface is designed to be simple, accessible, and multilingual to ensure that users from diverse backgrounds can easily understand and act upon disaster information.

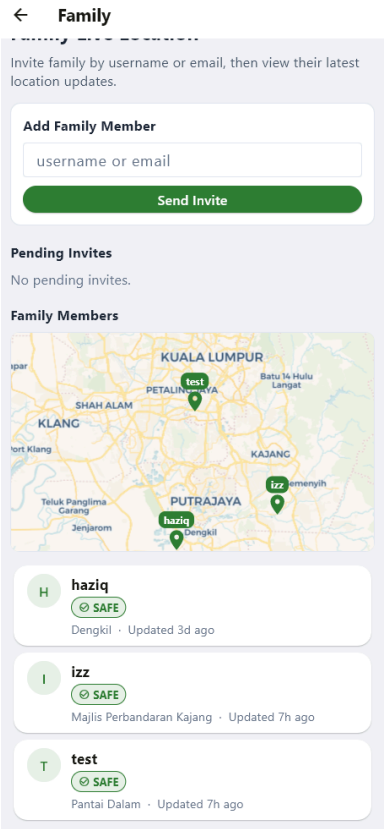
The following screenshots illustrate the key features of the platform, highlighting how users interact with the system during different stages of disaster preparedness and response.

No	Key Features	Description

1.		**<u>Preparedness Learning Module</u>** This interface provides educational modules designed to improve disaster awareness and preparedness among community members. Users can access step-by-step guidance on evacuation procedures, emergency planning, and safety practices before a disaster occurs. The learning module aims to promote **proactive disaster preparedness**, ensuring that users understand the necessary actions to take when receiving an early warning
2.		**<u>Hyper-Local Early Warning Notification</u>** This feature delivers **location-based disaster alerts** directly to users. Alerts are categorized by severity levels such as advisory, warning, and evacuation alerts. The notification system ensures that warnings are clear, actionable, and easily understood by users. To maintain accessibility during connectivity disruptions, the platform also supports **SMS-based alert delivery** for areas with limited internet access.

3.		**<u>Risk Map and Safe Evacuation Routes</u>** The risk mapping feature provides an interactive map displaying **hazard zones, flood risk levels, and recommended evacuation routes**. Users can quickly identify safe areas and the nearest evacuation shelters based on their current location. This visual guidance helps communities make faster and safer decisions during emergency situations.

4.		**<u>Community Intelligence Reporting System</u>** The platform allows users to submit **real-time reports about local disaster conditions**, such as rising water levels, blocked roads, or damaged infrastructure. These reports contribute to a community-driven information network that enhances situational awareness. The system aggregates and analyzes these reports to help authorities and communities identify emerging risks and coordinate response efforts.

5.		**<u>Family Safety Coordination Interface</u>** This feature allows families to remain connected during disaster situations. Users can share their safety status, track the location of family members, and coordinate evacuation plans. The goal of this feature is to reduce panic and improve communication among family members during emergencies.

6. AI Acknowledgement

The development of Disaster Resilience AI was supported by a range of AI tools, each serving a distinct and purposeful role throughout the project lifecycle. The team is committed to full transparency in disclosing these tools, consistent with responsible and ethical AI-assisted development practice.

6.1. Claude

Claude served as the team's primary AI assistant throughout the development process. It was used extensively to support code structuring across all three components of the platform which are the Flutter mobile frontend, the FastAPI backend, and the Python-based AI/ML model package. Specifically, Claude assisted with structuring API endpoint logic, working through Flutter widget hierarchies, resolving integration issues between Supabase and backend services, and explaining debugging approaches when the team encountered runtime errors or unexpected behaviours. Notably, the platform itself integrates Claude as a live feature which is a multi-agent analysis service powered by the Anthropic API performs contextual validation of community disaster reports by cross-referencing weather data and government alerts in real time. Throughout the project, Claude functioned as a collaborative pair-programming partner that helped the team work faster and more confidently which allowed all final decisions, architecture choices, and implementations to remain with the team.

6.2. Google Gemini

Google Gemini was used in two capacities. First, it supported the team's early research phase by providing comparative analysis of existing disaster management solutions and platforms, helping the team identify gaps that Disaster Resilience AI could meaningfully address. Second, Gemini was used to generate the project's logo and associated visual assets, giving the application a consistent and professional visual identity.

6.3. Kiro AI

Kiro AI was used during the early planning phase of the project for requirements gathering. The tool helped the team structure and articulate functional and non-functional requirements, translating the project's broad goals which are community reporting, family safety coordination, preparedness education, and hyper-local alerting, thus turning it into clearly defined specifications. This grounded the team's development priorities and helped maintain scope discipline throughout the hackathon.

6.4. Cursor & Antigravity

Cursor and Antigravity, are AI-enhanced code editors, and were used as part of the team's development environment. Its AI-assisted code navigation and inline suggestions complemented the team's workflow when working across the multi-component monorepo, making it easier to navigate between the Flutter, FastAPI, and ML model codebases efficiently.

6.4. Perplexity AI

Perplexity AI was used as a research assistant to gather up-to-date references and contextual information during both the development and documentation phases. It supported the team in drafting and refining sections of the project report by surfacing relevant background information quickly, enabling the team to write with greater accuracy and confidence.

6.5. Summary

Across all tools, the team's approach was consistent where AI was used to enhance productivity and reduce friction, not to replace critical thinking or decision-making. Every architectural decision, feature design, and implementation choice was made and reviewed by the team.

7. Conclusion

This project presents the LANDA Disaster Resilience AI platform, an integrated solution designed to improve disaster preparedness, early warning dissemination, and community coordination in flood-prone regions across Southeast Asia. By combining predictive analytics, hyper-local warning systems, community reporting, and accessible communication channels, the platform addresses critical gaps in existing disaster early-warning systems, particularly the lack of localized alerts, inclusive communication, and real-time community intelligence.

The proposed system demonstrates how digital technologies and artificial intelligence can support more effective disaster risk reduction. Through features such as interactive risk maps, evacuation guidance, multilingual alerts, and community-based reporting, the platform aims to transform disaster information into clear, actionable guidance for communities. The prototype illustrates the feasibility of integrating these components within a scalable architecture that supports continuous learning and improved forecasting over time.

In alignment with Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities, the platform contributes to strengthening disaster resilience by enhancing early warning coverage, improving evacuation readiness, and promoting inclusive communication for vulnerable populations.

Despite these promising outcomes, further development is necessary to fully implement and deploy the system in real-world environments. Future work may include integrating additional real-time data sources, improving the accuracy of predictive models through larger datasets, expanding multilingual support for more regional languages, and strengthening partnerships with local authorities and disaster management agencies. Field testing and pilot deployments in selected communities would also be valuable to evaluate usability, reliability, and social impact.

Overall, the Disaster Resilience AI platform demonstrates the potential of combining artificial intelligence, community participation, and accessible design to create a more responsive and inclusive disaster management ecosystem. By empowering communities with timely information and practical tools, the system aims to support safer, more resilient societies in the face of increasing climate-related disasters.

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